RAJALAKSHMI ENGINEERING COLLEGE

Harikumar K 251801086-2025-AIDS

H2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 10:50 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 10:52 AM |
| **Time taken** | 2 mins 36 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int findFirstZero(int a[], int low, int high) {
    if (low > high)
        return -1;

    int mid = low + (high - low) / 2;

    if (a[mid] == 0 && (mid == 0 || a[mid - 1] == 1))
        return mid;

    if (a[mid] == 1)
        return findFirstZero(a, mid + 1, high);
    else
        return findFirstZero(a, low, mid - 1);
}

int main() {
    int m;
    scanf("%d", &m);

    int a[m];

    for (int i = 0; i < m; i++) {
        scanf("%d", &a[i]);
    }

    int firstZero = findFirstZero(a, 0, m - 1);

    if (firstZero == -1)
        printf("0\n");
    else
        printf("%d\n", m - firstZero);

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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H2

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|---|---|
| **Started on** | Thursday, 27 August 2026, 10:53 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 10:54 AM |
| **Time taken** | 1 min 3 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

**Example 1:**

```
Input: nums = [3,2,3]
Output: 3
```

**Example 2:**

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

**Constraints:**

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

**For example:**

| Input | Result |
|---|---|
| 3<br>3 2 3 | 3 |
| 7<br>2 2 1 1 1 2 2 | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int majorityElement(int nums[], int n) {
    int candidate = nums[0];
    int count = 1;

    for (int i = 1; i < n; i++) {
        if (nums[i] == candidate)
            count++;
        else
            count--;

        if (count == 0) {
            candidate = nums[i];
            count = 1;
        }
    }

    return candidate;
}

int main() {
    int n;
    scanf("%d", &n);

    int nums[n];
```

```
27
28      for (int i = 0; i < n; i++) {
29          scanf("%d", &nums[i]);
30      }
31
32      printf("%d\n", majorityElement(nums, n));
33
34      return 0;
35  }
36
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 3<br>3 2 3 | 3 | 3 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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H2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 10:54 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 10:56 AM |
| **Time taken** | 2 mins 13 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

**Output Format**

First Line Contains Integer – Floor value for x

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int findFloor(int a[], int low, int high, int x) {
    if (low > high)
        return -1;

    int mid = low + (high - low) / 2;

    if (a[mid] == x)
        return a[mid];

    if (a[mid] > x)
        return findFloor(a, low, mid - 1, x);

    int result = findFloor(a, mid + 1, high, x);

    if (result == -1)
        return a[mid];

    return result;
}

int main() {
    int n, x;

    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &a[i]);

    scanf("%d", &x);

    printf("%d\n", findFloor(a, 0, n - 1, x));

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 6<br>1<br>2<br>8<br>10<br>12<br>19<br>5 | 2 | 2 | ✔ |
| ✔ | 5<br>10<br>22<br>85<br>108<br>129<br>100 | 85 | 85 | ✔ |
| ✔ | 7<br>3<br>5<br>7<br>9<br>11<br>13<br>15<br>10 | 9 | 9 | ✔ |

Passed all tests! ✔

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Marks for this submission: 1.00/1.00.

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H2

| | |
|---|---|
| **Started on** | Thursday, 27 August 2026, 10:56 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 10:58 AM |
| **Time taken** | 1 min 26 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement:**

Given a sorted array of integers say arr[] and a number x. Write a recursive program using divide and conquer strategy to check if there exist two elements in the array whose sum = x. If there exist such two elements then return the numbers, otherwise print as "No".

Note: Write a Divide and Conquer Solution

**Input Format**

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Sum Value

**Output Format**

First Line Contains Integer – Element1

Second Line Contains Integer – Element2 (Element 1 and Elements 2 together sums to value "x")

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int binarySearch(int arr[], int low, int high, int key)
{
    if (low > high)
        return -1;

    int mid = low + (high - low) / 2;

    if (arr[mid] == key)
        return mid;

    if (arr[mid] > key)
        return binarySearch(arr, low, mid - 1, key);

    return binarySearch(arr, mid + 1, high, key);
}

int findPair(int arr[], int left, int right, int n, int x,
             int *element1, int *element2)
{
    if (left > right)
        return 0;

    int mid = left + (right - left) / 2;
    int required = x - arr[mid];

    int pos = binarySearch(arr, 0, mid - 1, required);

    if (pos == -1)
        pos = binarySearch(arr, mid + 1, n - 1, required);

    if (pos != -1)
    {
        *element1 = arr[mid];
        *element2 = arr[pos];
        return 1;
    }

    if (findPair(arr, left, mid - 1, n, x, element1, element2))
        return 1;

    return findPair(arr, mid + 1, right, n, x, element1, element2);
}

int main()
{
    int n;
    scanf("%d", &n);

```

```
51    int arr[n];
52
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 4<br>2<br>4<br>8<br>10<br>14 | 4<br>10 | 4<br>10 | ✔ |
| ✔ | 5<br>2<br>4<br>6<br>8<br>10<br>100 | No | No | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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|---|---|
| **Started on** | Thursday, 27 August 2026, 10:58 AM |
| **State** | Finished |
| **Completed on** | Thursday, 27 August 2026, 10:59 AM |
| **Time taken** | 42 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Write a Program to Implement the Quick Sort Algorithm

Input Format:
The first line contains the no of elements in the list-n
The next n lines contain the elements.

Output:
Sorted list of elements

**For example:**

| Input | Result |
|---|---|
| 5<br>67 34 12 98 78 | 12 34 67 78 98 |

**Answer:**

```c
#include <stdio.h>

void quickSort(int arr[], int low, int high)
{
    if (low < high)
    {
        int i = low;
        int j = high;
        int pivot = arr[low];

        while (i < j)
        {
            while (arr[i] <= pivot && i < high)
                i++;

            while (arr[j] > pivot)
                j--;

            if (i < j)
            {
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }

        arr[low] = arr[j];
        arr[j] = pivot;

        quickSort(arr, low, j - 1);
        quickSort(arr, j + 1, high);
    }
}

int main()
{
    int n;
    scanf("%d", &n);

    int arr[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &arr[i]);

    quickSort(arr, 0, n - 1);

    for (int i = 0; i < n; i++)
    {
        printf("%d", arr[i]);
        if (i < n - 1)
            printf(" ");
```

```
52 |     }
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>67 34 12 98 78 | 12 34 67 78 98 | 12 34 67 78 98 | ✔ |
| ✔ | 10<br>1 56 78 90 32 56 11 10 90 114 | 1 10 11 32 56 56 78 90 90 114 | 1 10 11 32 56 56 78 90 90 114 | ✔ |
| ✔ | 12<br>9 8 7 6 5 4 3 2 1 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | 1 2 3 4 5 6 7 8 9 10 11 90 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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